AI-Assisted Design of Experiments (DOE) Optimization System

■ What Chemists Actually Do (Reality)

1. Work with multiple variables such as Temperature, Pressure, Solvent, Catalyst, and Time
2. Traditional approach involves manual experimentation with combinations, which is slow and inefficient
3. DOE approach uses structured experimental design to reduce trials, identify key factors, and optimize yield

■■ Core Concepts

1. Factorial Design: Tests all combinations of variables (e.g., 2 levels × 3 factors = 8 experiments)
2. Fractional Factorial Design: Uses a subset of experiments to significantly reduce trials
3. Response Surface Methodology (RSM): Builds models to find optimal conditions
4. Regression Models: Predict outcomes such as yield and purity

■ Project Objective

1. Develop an AI-assisted DOE system
2. Use historical experimental data as input
3. Suggest minimum number of experiments
4. Predict outcomes (yield, purity)
5. Optimize process conditions

■■ System Architecture

1. Input Layer: Historical experimental data (e.g., Temp, Pressure, Catalyst, Yield)
2. DOE Engine: Implement factorial/fractional design and RSM using pyDOE2
3. Modeling Layer: Use scikit-learn for regression models
4. AI Layer: Use LLMs to explain results and suggest next experiments
5. Optimization Engine: Use Bayesian optimization or gradient methods
6. Output Layer: Suggested experiments, predicted results, and visualization graphs

■■ UI Design

1. Screen 1: Upload experimental data
2. Screen 2: Select variables and define ranges
3. Screen 3: View suggested experiments and optimization results

■ Advantages Over Traditional Tools

1. Includes AI-based suggestions
2. Supports continuous learning from new data

3. Integrates with pharma-specific workflows
4. Provides automated insights beyond statistical outputs

■ Agentic AI Components

1. Data Agent: Cleans and prepares data
2. Modeling Agent: Builds predictive models
3. Insight Agent: Interprets results
4. Optimization Agent: Suggests next experiments

■ Interview Question

1. Design an AI system to reduce experimental trials using DOE and historical data
2. Explain use of statistical methods
3. Explain role of AI enhancement
4. Explain how next experiments are suggested

■ Expected Strong Answer

1. Use factorial or fractional DOE
2. Build regression models
3. Apply optimization techniques like Bayesian optimization
4. Use LLMs for interpretation
5. Suggest next best experiments

■■ Important Note

1. Do not attempt to fully replicate complex tools like Minitab initially
2. Start with simple factorial design and regression models
3. Gradually integrate AI and optimization features